# How far ahead do public data forecast a large dengue season? Predictors and lead time by Brazilian state

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**Abstract.** Background: the number of Brazilian states with a high-incidence dengue season (300 or more probable cases per 100,000) ranged from 3 to 19 between 2014/15 and 2024/25. It is not known how far ahead of the simple case count public data allow forecasting this. Methods: each state had its own season, starting at its week of lowest incidence, and all counts used probable cases only. For the 27 states and 11 seasons (297 pairs), at each of the first 43 weeks of the season, rebuilding what was available on each date (6-week lag of the open SINAN file, data-entry delay, and past outcomes only once known), six predictors (count projection, recent growth, off-season cases, mild winter, thermal suitability and El Niño) were tested alone and in pairs, in penalized logistic regressions anchored on the state's history. Criterion M is the model with the lowest error in earlier seasons, chosen without seeing the forecast season. Lead time is the number of weeks between accuracy stably passing 75% and the count of probable cases making half of the high seasons evident. Uncertainty comes from 4,000 resamples by season. Results: off-season cases were the predictor that added most to the count projection (incremental BSS 0.077; 95% CI 0.029 to 0.131), and M became projection plus off-season cases. M passed 75% accuracy at week 18 (75.1%; CI 68.7% to 81.5%; sensitivity 75.7%), and half of the high seasons became evident at week 35: a lead time of 17 weeks (CI 10 to 25), or 9 under the strict rule. The projection alone passed 75% at week 10, with 25 weeks of lead time (CI 14 to 30; 11 under the strict rule), and was as accurate as M or more in almost every week. Without the publication lag, lead times were 18 and 25 weeks. Conclusion: with criterion M, made up of the count projection and off-season cases, a high season can be forecast 17 weeks ahead of the simple case count (9 under the strict rule), in this retrospective reconstruction; the projection alone anticipates even more, and M showed no clear advantage over it.

**Keywords:** dengue; forecasting; lead time; predictors; SINAN; Brazilian states.

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*Translated from the Portuguese original.*

**In plain language**

*Why did we do this study?* Some years have a lot of dengue and others have little. People who run health services in each state need to know early whether the year will be bad. We wanted to know how much spare time the data the government publishes give before simply adding up the cases makes everything obvious.

*What did we do and what did we find?* We went back over eleven past years and pretended to be in each week, using only what was already known. We tested six clues, such as heat, winter, El Niño and cases that show up outside the usual season. The most useful clue was the running total of cases itself, compared with each state's usual pace, and the one that helped most alongside it was the cases outside the usual season. With these clues, on average, the answer was right in three out of four forecasts some months before the result became obvious. Some years it did better, some worse.

*What does this mean?* Public data already give an early warning, but the test looked back in time. It still has to be seen whether it works in the coming years. This note says nothing about the health of any person.

## I. INTRODUCTION

### A. The problem and why it matters

Dengue in Brazil is seasonal, but the size of each season varies greatly from one year to the next. Between 2014/15 and 2024/25, the number of states with a high-incidence season ranged from 3 to 19, out of 27 (Table II). For those who organize beds, staff, supplies and campaigns, the practical question is: how far ahead do public data make it possible to say that this year will be a big one in my state? And is that lead time greater than what the case count itself would already give?

The idea of this note is that of an election vote count. Each week, more data arrive. At the start, the count says little; at the end, the case count already shows the result without any model at all. A forecasting model is only useful if it gets it right before that moment. The distance, in weeks, between the point at which the model starts to get it right and the point at which the count makes the result evident is the **lead time** that this note measures. It is not, in itself, the lead time relative to the start of the epidemic or to the peak; for that reason, the note also measures how many high seasons the model anticipates before the peak.

### B. What is already known

Forecasting dengue is hard, and the literature is candid about it. In an open challenge with 16 teams, eight seasons and two cities in the Americas, forecasts made at the start of the season had little skill, skill was lowest precisely in the high-incidence seasons, and the models with more biology were not, on average, better than the simple ones [5]. In Vietnam, an ensemble of climate-driven models outperformed a historical baseline at 1 to 3 months ahead, but not at 4 to 6 [6]. In Brazil, seasonal climate forecasts made it possible to issue warnings three months before the 2014 World Cup [3], and the evaluation of that prototype used historical averages as the comparison [4]. The InfoDengue system has estimated the cases not yet reported every week (nowcasting) since 2015 [2]; the InfoDengue-Mosqlimate consortium brings teams together to forecast each state's incidence in the following season [7, 8]; and there are studies that combine official data and internet searches to estimate the current week [9].

These works were located through a general internet search and by reading the full text in Europe PMC, on 7 Nov 2026. There was no systematic search of the literature (PubMed, SciELO, LILACS and other databases were not searched systematically); for that reason, this note does not claim that the approach is new.

### C. The question

This note asks, using the aggregated public data from SINAN:

- **Predictors:** which signals, alone and in pairs, improve the forecast of a high-incidence season relative to the state's history, and how much does each one add to the count projection?
- **Start:** from which week of the season does the chosen forecasting criterion get the yes-or-no forecast right in 75% or more of the state-season pairs, without falling below that again?
- **Lead time:** how many weeks separate that start from the week in which the simple case count of the probable cases already available makes half of the high seasons evident?

**Why 75% accuracy?** It is the level at which the forecast is wrong, on average, in at most one out of every four state-season pairs. It is above the accuracy of someone who always answers "no" (54% in the period, because 46% of the seasons were high) and the accuracy of someone who only looks at the state's history (64% to 66%). It is a usefulness threshold, set in the protocol, and not a significance test. Overall accuracy is not the same as identifying 75% of the high seasons; for that reason the note also reports the sensitivity (fraction of the high seasons identified), the specificity and the accuracy in each season.

### D. What this note does and does not do

The note evaluates, over the 11 seasons from 2014/15 to 2024/25 and the 27 states, six predictors alone and in pairs, a forecasting criterion chosen among them and two baselines, in each of the first 43 weeks of each state's season, reconstructing what would have been available at each date. It also records a forecast for 2026/27, to be checked. It does not forecast the number of cases or the date of the peak, does not work with municipalities and does not provide guidance to people or services.

## II. DATA

**Unit.** Each of the 27 states (26 states and the Federal District).

**Only probable cases, in all years.** The open files for 2014, 2015 and 2022 to 2026 contain only the notifications that were not discarded; those for the other years also contain the discarded ones. So that the series is the same in all years, all counts in this note use only probable cases (notifications not discarded): the outcome, the projection, the growth, the off-season cases and the evidence rule. There are about 24.3 million probable cases from 2007 to 2026. This choice uses the final classification of each notification, which did not exist in real time; anyone following the system also sees notifications that will later be discarded.

**One season for each state.** The dengue calendar is not the same across the whole country. In the South, Southeast and Center-West, the peak usually comes between February and April; in part of the Northeast, between May and July; in some states in the North, at the end of the year. A single calendar, from July to June, would cut some states' epidemics in half. For that reason each state has its own season, which begins at its **trough week**: the epidemiological week with the lowest mean incidence, calculated with the years 2007 to 2013, prior to the seasons evaluated. The season lasts one year from the trough. In most states, the trough falls between late August and mid-October (weeks 35 to 42); in Alagoas and Paraíba, around the turn of the year; in Rio Grande do Sul, in July (Figure 1).

**Outcome.** A season is high-incidence when the probable cases with symptom onset in the state's season add up to 300 or more per 100,000 inhabitants, the threshold used in surveillance bulletins for high incidence. A total of 297 state-season pairs were evaluated, of which 136 (45.8%) were high-incidence. Counted from the trough, the median week of the peak of the high seasons was week 28 (half between weeks 23 and 33).

**What was known each week.** The forecast made on date D should use only what was available on D. Three delays were reconstructed. The first is the **data entry** delay: a case only counts if it was entered before the cutoff. In the files without a data entry date (2016 to 2020) and in the records with an invalid date, the probability that each case was already in the system was estimated from the delay distribution of the same state and the same month in years earlier than or equal to that of the file (2013 to 2015 for the files from 2016

Table I. Public data sources used, with period and main limitation.

| Source | What it provides | Period | Main limitation |
|---|---|---|---|
| SINAN dengue (OpenDataSUS) [15] | de-identified individual notifications, with date of symptom onset, of notification, of data entry and final classification | files from 2007 to 2026, downloaded on 7 Nov 2026 | no data entry date from 2016 to 2020; without the discarded notifications in 2014, 2015 and 2022 to 2026; 2026 file up to the week of 27 Sep 2026 |
| IBGE [18] | population by state and year; population of the municipalities in the 2022 Census | 2006 to 2025 | 2007 and 2023 interpolated |
| NASA POWER (MERRA-2) [16] | daily mean temperature | 2006 to Nov 2026 | gridded reanalysis, not station data; 6 points per state; version of 7 Nov 2026 |
| NOAA, ONI index [17] | Pacific temperature anomaly (El Niño) | 1950 to Sep 2026 | a single number for the whole planet |

Table II. Number of states, out of 27, with a high-incidence season in each season evaluated.

| Season | States with high incidence | Season | States with high incidence |
|---|---|---|---|
| 2014/15 | 17 | 2020/21 | 10 |
| 2015/16 | 18 | 2021/22 | 17 |
| 2016/17 | 3 | 2022/23 | 13 |
| 2017/18 | 3 | 2023/24 | 19 |
| 2018/19 | 14 | 2024/25 | 12 |
| 2019/20 | 10 | | |

onward). The second is the **publication** delay: the 2026 file, downloaded on 7 Nov 2026, only has complete weeks up to the one beginning on 20 Sep 2026, about 6 weeks earlier; for that reason, the forecast made on date D uses only cases entered up to D minus 6 weeks. The third is the **outcome** delay: the result of a past season is only treated as known 8 weeks after the end of the season plus the publication lag, and this rule applies to the history, to the projection, to the training of the models and to the choice of the criterion. Temperature enters with a 1-week delay and the El Niño index with 2 months. The reconstruction is approximate: the final classification of the cases, the temperature series downloaded today (which may have been revised after the simulated dates [16]) and, in the records without a data entry date, the delay distribution from other years are not exactly what was seen at the time.

## III. METHODS

### A. Overview of the design

The study simulates what an analyst would have done in each week of the past. For each state, each season from 2014/15 to 2024/25 and each week k of the season, from k = 0 (the trough) to k = 42, the models are trained only on season pairs whose outcome was already known at the date, in the same week k; the forecast is then compared with what happened. Because the states have different calendars, the training for each season and week uses only the outcomes known at the earliest date among the forecasts for that week, which is conservative for the states that start later. That gives 43 weeks $\times$ 11 seasons $\times$ 27 states = 12,771 forecasts per model. The protocol was registered by a dated commit (a9c960e, 7 Nov 2026), before the commit of the results; this record documents the order, but does not prove that the data had not been explored before, and the retrospective evaluation is not blind.

### B. Two baselines

**R0, the history.** It is the answer of someone who does not look at the season's data: the fraction of high seasons the state had in the past, among those with a known outcome, with a small (Laplace) correction so as not to give 0% or 100%. For a state that had 13 high seasons out of 19, for example, R0 = (13 + 1)/(19 + 2) = 67%.

**A, the count projection.** It is the answer of someone who looks at the count and applies a rule of three. It divides the probable cases already available by the fraction that, in the previous seasons of the same state with a known outcome, was already available in the same week relative to the final total. If, in previous years, on average 20% of the season's cases had arrived by this week, and now there are 100 cases per 100,000, the projection is 500 per 100,000:

$$\widehat{I} = \frac{N_{\text{disp}}(k)}{g(k)} \times \frac{100\,000}{\text{pop}} \tag{1}$$

where $N_{\text{disp}}(k)$ is the number of probable cases available in week k and g(k) is the median, over the state's previous seasons (at least 3), of the fraction available in that week. The probability of a high season comes from a simple logistic regression on the logarithm of the projection. In the first weeks, when the lag has not yet let any data from the season arrive, A has no information of its own.

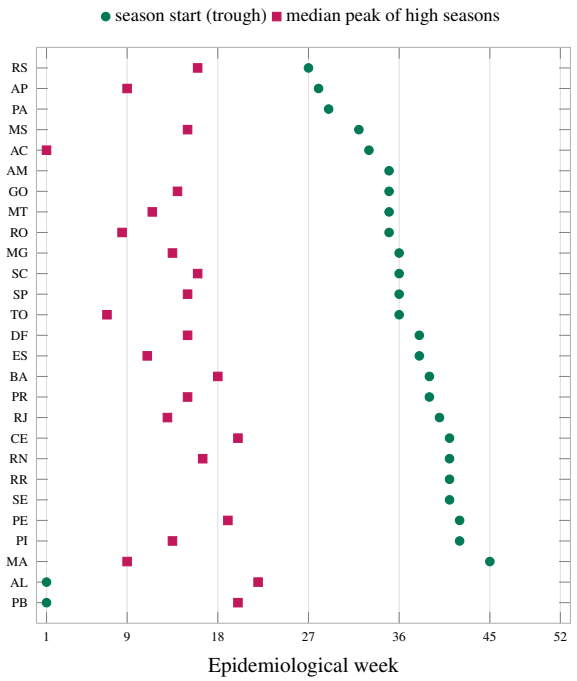



Fig. 1. Epidemiological week of the start of the season (trough) of each state, calculated with the years 2007 to 2013, and median week of the weekly peak of the high-incidence seasons from 2014/15 to 2024/25. States with no high season in the period appear only with the trough.

### C. Six candidate predictors

**What is thermal suitability?** The *Aedes aegypti* mosquito and the virus depend on temperature in almost everything: how much the mosquito bites, how long it lives, how long it takes to become an adult and how long the virus takes to reach the mosquito's saliva. Mordecai and colleagues combined these traits, measured in the laboratory, into a mechanistic model of transmission potential as a function of temperature, which is zero below 17.8 °C and above 34.6 °C and peaks at 29.1 °C [1]. This note does not reproduce that model: it uses a simple unimodal curve, built for the study, that respects these three values (Figure 2). It was rescaled from 0 to 1 and calculated for each day from the mean temperature of 6 points per state (the capital and the 5 most populous municipalities in the 2022 Census), weighted by population [16, 18, 19]. The original article distinguishes constant temperature from temperature that varies over the day; here, only the daily mean is used. The curve does not measure how many mosquitoes there are, only how much the climate favors transmission.

$$f(T) = \left(\frac{T - 17.8}{11.3}\right)^{\alpha} \left(\frac{34.6 - T}{5.5}\right)^{\beta} \qquad (2)$$

for T between 17.8 °C and 34.6 °C, and zero outside that range, with $\alpha + \beta = 3$ and $\alpha = 3 \times 11.3/16.8$, which places the maximum, equal to 1, at 29.1 °C.

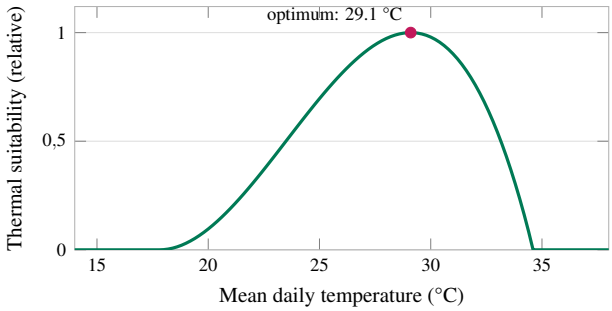



Fig. 2. Thermal suitability curve used in this note: a unimodal approximation built from the limits (17.8 °C and 34.6 °C) and the optimum (29.1 °C) of dengue transmission by *Aedes aegypti* published by Mordecai et al.; it is not the article's mechanistic model.

### D. Screening: each predictor alone and in pairs

Each predictor alone (6 models) and each pair (15 models) enters a logistic regression that starts from the state's history and adjusts it with the signals:

$$\operatorname{logit} p = \operatorname{logit} R_0 + \beta_0 + \beta_1 x_1 + \beta_2 x_2 \qquad (3)$$

Here, logit is the odds scale (logit p = ln[p/(1 - p)]) and x1, x2 are the predictors standardized in training. A penalization pulls the weights $\beta$ toward zero, so that the model only moves away from the history when the data give sufficient evidence; its strength is chosen among five values by the lowest error in the last three seasons of each training set. The screening compares the 21 models by the **Brier score** [10], the mean squared error of the probability, summed over all weeks, and by the gain over a baseline, BSS = 1 - Brier(model)/Brier(baseline): positive when the model makes smaller errors. The **incremental gain** of each second predictor is the BSS of the pair with the projection over the projection alone, in the same model: this isolates the effect of the second predictor.

### E. Criterion M

Criterion M is the screening model that best forecast the past seasons. To forecast season t, the one among the 21 with the lowest Brier score in the forecasts of the seasons from 2011/12 to t - 2 is chosen, all with an outcome known before any forecast of t. The accuracy of M measures the performance of the whole procedure, including the model changes from one season to another. The composition of M stated in the note is the one chosen with all seasons up to 2024/25. Projection A and criterion M do not have the same form (A does not start from the history and has a fixed penalization); for that reason, the comparison between M and A does not isolate the effect of any predictor, and the gain of each predictor is measured by the incremental gain.

Table III. Candidate predictors and the hypothesis behind each one. All measured by state and by forecast date, using only what was available.

| Predictor | How it is measured | Hypothesis |
|---|---|---|
| Count projection | logarithm of the incidence projected by the rule of three above | what has already arrived, corrected for the state's typical pace, anticipates the total |
| Recent growth | slope of the logarithm of weekly cases over the last 4 available weeks | a curve that accelerates tends to end high |
| Off-season cases | logarithm of the incidence with symptom onset in the first 13 weeks of the season (the low season), minus the median of the same value, in the same week, in the state's previous seasons | above-normal transmission when the climate is unfavorable indicates sustained circulation of virus and mosquito, which is amplified when the heat arrives |
| Mild winter | anomaly of the mean temperature over the state's 91 climatologically coldest days, within the previous 365 days, relative to the 2006 to 2013 mean | a warmer winter leaves the mosquito and transmission uninterrupted |
| Thermal suitability | mean, over the previous 8 weeks, of a temperature-dependent transmission curve (below) [1] | favorable temperature speeds up the cycle of the mosquito and the virus |
| El Niño (ONI) | last available value of the ONI index [17] | El Niño alters heat and rainfall in Brazil and has already been used in dengue warnings [3] |

### F. How accuracy is measured

Each model gives a probability. When it is 50% or more, the forecast is "high season"; below that, "not high". **Accuracy** is the fraction of state-season pairs in which the forecast matched what happened; **sensitivity** is the fraction of high seasons with a "high" forecast; **specificity**, the fraction of non-high seasons with a "not high" forecast.

**Example**

In any given week, if the model said "high" for 12 states and "not high" for 15, and got 9 of the 12 and 12 of the 15 right, the season's accuracy is (9 + 12)/27 = 78%. If 11 states had a high season and the model caught 9 of them, the sensitivity is 9/11 = 82%. The week's accuracy is the same calculation summed over the 11 seasons, over 297 pairs.

### G. Uncertainty, chance and margin of error

Eleven seasons are few, and the accuracy measured in them varies by chance. To measure this uncertainty, the whole calculation was redone 4,000 times, drawing with replacement 11 seasons from the 11 available (bootstrap [11]). Whole seasons are drawn, not states, because the states are not independent within the same year: when there is a national epidemic, many rise together. Two measures come out of this:

- **95% interval:** the range containing 95% of the resampled values; it indicates, approximately, between which values the true value probably lies.
- **Chance of accuracy of 75% or more:** the fraction of resamples in which accuracy was 75% or more,

$$c = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left[\text{acerto}_b^* \geq 0.75\right], \qquad B = 4\,000 \tag{4}$$

where $\mathbf{1}[\cdot]$ equals 1 when the condition is true and 0 when it is not. The margin of error of the draw (Monte Carlo) is $\pm 1.96\sqrt{c(1-c)/B}$, at most ±1.5 percentage points with B = 4,000. The uncertainty that comes from having only 11 seasons does not decrease with more draws, and it is the one the 95% interval shows. For that reason c is an approximate measure of confidence, and not the exact probability that the true accuracy exceeds 75%. In the screening, the corresponding chance is the fraction of resamples with a positive gain. Following the program's rule, chances of 75% or more are highlighted as promising.

### H. Start, evidence and lead time

- **Start (main rule):** the first week k from which accuracy is 75% or more in it and in all following weeks up to k = 42.
- **Start (strict rule):** the same, requiring the lower limit of the 95% interval of accuracy to be 75% or more. Both are reported, because the lead time depends on which one is used.
- **Evident from the count:** a season is evident in week k when the probable cases already available exceed 300 per 100,000. Because only probable cases are counted, evidence implies a high season. At that point no model is needed: counting is enough.
- **Window lead time:** the number of weeks between the start and the first week in which half or more of the high seasons are already evident. If the start comes later, the lead time is zero. The intervals include resamples without a window as zero, and the note reports their fraction.
- **Lead time per season:** for each high season, the week in which the count makes it evident minus the first week from which the model says "high" without interruption

up to it; and whether that first week came before the peak. High seasons that do not become evident by week 42 enter with evidence at week 43 (censoring). It includes the states in which the model says "high" from the start based on the history alone, and for that reason it comes with the false alarm rate: the fraction of non-high seasons that the model was calling "high" in the start week.

**Why is there no correction for multiple testing?** The main question is not a hypothesis test in each week, but the reading of an accuracy curve against a threshold. Looking at 43 weeks, however, increases the chance that one of them exceeds 75% by chance. The protection adopted is the stability rule and the strict rule. In the screening, 21 models were compared; nested selection prevents the luck of any one of them in the evaluated seasons from entering the accuracy of M.

### I. Sensitivity analyses

Seven variations registered in the protocol: publication lag of 0 weeks (as someone accessing the system without waiting for the open file would have) and of 10 weeks; evidence with 25% and with 75% of the high seasons; exclusion of the atypical seasons from 2015/16 to 2017/18 (Zika and chikungunya circulation) and 2019/20 to 2020/21 (COVID-19 pandemic), also from the calculation of the predictors and the history; and a national season, from July to June, for all states.

**Protocol deviations.** None.

## IV. RESULTS

**How to read these results**

- Weeks are counted from the start of each state's season. For São Paulo, for example, week 0 is early September, week 10 is mid-November, week 18 is mid-January, week 26 is early March and week 35 is mid-May.
- Accuracy is the fraction of state-season pairs with the correct yes-or-no forecast; the threshold is 75%, and the history alone gets 64% to 66% right.
- The 95% interval shows the uncertainty; the chance is the fraction of resamples that exceed the threshold, in bold when it reaches 75%.
- Lead time is the number of weeks between the model starting to be right 75% or more of the time and the count of probable cases making half of the high seasons evident.

### A. Screening of the predictors

Alone, two predictors greatly reduced the error of the history: the count projection (BSS 0.298; 95% CI 0.245 to 0.351) and **off-season cases** (0.278; CI 0.186 to 0.372), both with a chance of 100%. Thermal suitability, mild winter, El Niño and recent growth, alone, were close to zero. The best pair was **projection and off-season cases** (BSS 0.352 over the history). Added to the projection, in the same model, off-season cases contributed a BSS of 0.077 (CI 0.029 to 0.131; chance of 100%) and thermal suitability 0.027 (CI 0.004 to 0.056; chance of 99%); El Niño (0.004; chance of 57%), mild winter and growth added no clear gain (Table IV). Against projection A, the pair projection and off-season cases had a BSS of 0.037 (CI -0.010 to 0.088; chance of 94%).

In the nested selection, M was the projection alone in 4 seasons (2015/16 to 2018/19), projection and El Niño in 3 (2019/20 to 2021/22), **projection and off-season cases** in the last 3 (2022/23 to 2024/25) and projection and mild winter in the first. With all seasons up to 2024/25, the one chosen is **projection and off-season cases**: this is the composition of criterion M from now on. The accuracy of M reported below is that of the procedure, with the model changes over time.

### B. Week-by-week accuracy and lead time

With the 6-week lag of the open file, M exceeded 75% accuracy in **week 18** (75.1%; 95% CI 68.7% to 81.5%; chance of 52%) and stayed above it until the end; under the strict rule, in week 26 (81.1%). Half of the high seasons became evident from the count in **week 35**. The lead time of M was **17 weeks** (95% CI 10 to 25) under the main rule and 9 under the strict rule; the window existed in all resamples. Projection A exceeded 75% in **week 10** (76.1%; CI 71.7% to 80.1%; chance of 70%), with **25 weeks** of lead time (CI 14 to 30); under the strict rule, in week 24, with 11 weeks (Tables V and VI and Figure 3). The median week of the peak of the high seasons was week 28: M opens 10 weeks before it, and A 18 weeks before it.

In accuracy, A was equal to or better than M in 33 of the 35 weeks from week 8 onward, and the probabilities of M were worse than those of A (BSS of -0.05 to -0.15 over A from week 10 onward). M showed no clear advantage over A in this experiment. Because the two do not have the same form, this does not mean that off-season cases do not help: in the same model, they improved the projection.

The 75% accuracy is an average. In the start week of M, it identified 75.7% of the high seasons and 74.5% of the non-high ones, and accuracy ranged from 59% to 96% across seasons; in that of A, the sensitivity was 70.6%, that is, almost 3 out of every 10 high seasons still had a "not high" forecast, and accuracy ranged from 63% to 85%.

**Per season.** Counted season by season, M started saying "high" without interruption, at the median, 24 weeks before the count made the season evident (CI 19 to 29); it anticipated 88% of the high seasons by 4 weeks or more and, in 77% of them (CI 68% to 87%), before the peak. This number is larger than the window's because it includes states in which M says "high" from the beginning based on the history, and it has a cost: in the start week, M was calling 25.5% of the non-high seasons "high". For A, the same numbers were 20 weeks (CI 15 to 22), 90%, 78% before the peak and 19.3% false alarm. In 34 of the 136 high seasons, the count did not reach 300 per 100,000 by

Table IV. Screening of the 21 models (6 predictors alone and 15 pairs), in order of gain over the history: BSS over R0 and incremental gain over the projection alone in the same model, summed over weeks 0 to 42 of the 11 seasons, with 95% CI and chance of a positive gain (4,000 resamples), and number of seasons in which the model was chosen as M. In bold, chance of 75% or more.

| Model | BSS over R0 (95% CI) | Chance | Gain over the projection (95% CI) | Chance | Chosen |
|---|---|---|---|---|---|
| projection + off-season cases | 0.352 (0.272 to 0.433) | **100%** | 0.077 (0.029 to 0.131) | **100%** | 3 |
| projection + thermal suitability | 0.317 (0.262 to 0.374) | **100%** | 0.027 (0.004 to 0.056) | **99%** | 0 |
| projection + El Niño | 0.300 (0.222 to 0.376) | **100%** | 0.004 (-0.050 to 0.061) | 57% | 3 |
| projection | 0.298 (0.245 to 0.351) | **100%** | | | 4 |
| projection + growth | 0.295 (0.241 to 0.350) | **100%** | -0.004 (-0.014 to 0.008) | 25% | 0 |
| projection + mild winter | 0.294 (0.233 to 0.358) | **100%** | -0.005 (-0.063 to 0.044) | 45% | 1 |
| off-season cases + mild winter | 0.283 (0.188 to 0.389) | **100%** | -0.020 (-0.104 to 0.079) | 32% | 0 |
| off-season cases + thermal suitability | 0.279 (0.185 to 0.373) | **100%** | -0.027 (-0.103 to 0.054) | 25% | 0 |
| off-season cases | 0.278 (0.186 to 0.372) | **100%** | -0.027 (-0.100 to 0.048) | 24% | 0 |
| off-season cases + El Niño | 0.273 (0.162 to 0.384) | **100%** | -0.036 (-0.119 to 0.055) | 21% | 0 |
| growth + off-season cases | 0.269 (0.174 to 0.365) | **100%** | -0.041 (-0.119 to 0.039) | 15% | 0 |
| mild winter + thermal suitability | 0.035 (-0.044 to 0.099) | **83%** | -0.374 (-0.519 to -0.242) | 0% | 0 |
| thermal suitability + El Niño | 0.016 (-0.042 to 0.065) | 72% | -0.402 (-0.495 to -0.314) | 0% | 0 |
| growth + thermal suitability | 0.010 (-0.034 to 0.052) | 66% | -0.410 (-0.545 to -0.298) | 0% | 0 |
| growth + mild winter | 0.008 (-0.051 to 0.057) | 63% | -0.412 (-0.542 to -0.292) | 0% | 0 |
| mild winter | 0.005 (-0.055 to 0.057) | 58% | -0.417 (-0.551 to -0.292) | 0% | 0 |
| thermal suitability | 0.002 (-0.045 to 0.047) | 54% | -0.421 (-0.557 to -0.303) | 0% | 0 |
| growth + El Niño | 0.000 (-0.046 to 0.045) | 50% | -0.424 (-0.517 to -0.343) | 0% | 0 |
| mild winter + El Niño | -0.003 (-0.064 to 0.054) | 48% | -0.428 (-0.535 to -0.324) | 0% | 0 |
| El Niño | -0.006 (-0.056 to 0.042) | 42% | -0.432 (-0.534 to -0.339) | 0% | 0 |
| growth | -0.018 (-0.047 to 0.009) | 11% | -0.449 (-0.572 to -0.344) | 0% | 0 |

Table V. Accuracy by week of the season with the 6-week lag of the open file (297 pairs per week): criterion M and projection A, with 95% CI and chance of accuracy of 75% or more, and fraction of the high seasons already evident from the count. In bold, accuracy or chance of 75% or more.

| Week | Accuracy M (95% CI) | Chance M | Accuracy A (95% CI) | Chance A | Evident |
|---|---|---|---|---|---|
| 0 | 60.6% (54.2% to 66.7%) | 0% | 49.5% (37.4% to 61.6%) | 0% | 0% |
| 8 | 64.6% (56.6% to 71.7%) | 0% | 69.4% (60.9% to 77.1%) | 9% | 0% |
| 10 | 68.7% (62.3% to 75.1%) | 3% | **76.1%** (71.7% to 80.1%) | 70% | 0% |
| 12 | 69.7% (63.0% to 76.4%) | 7% | **77.4%** (72.7% to 82.2%) | **86%** | 1% |
| 14 | 72.4% (66.0% to 79.1%) | 22% | **78.8%** (72.7% to 84.8%) | **88%** | 1% |
| 16 | 73.7% (67.3% to 80.5%) | 36% | **79.1%** (73.1% to 85.5%) | **90%** | 1% |
| 18 | **75.1%** (68.7% to 81.5%) | 52% | **79.5%** (73.7% to 85.5%) | **94%** | 1% |
| 21 | **77.1%** (71.0% to 83.2%) | **75%** | **80.1%** (74.7% to 85.5%) | **97%** | 4% |
| 24 | **79.1%** (72.7% to 84.8%) | **91%** | **82.8%** (77.4% to 88.2%) | **100%** | 8% |
| 26 | **81.1%** (76.1% to 85.9%) | **99%** | **83.8%** (78.8% to 88.9%) | **100%** | 11% |
| 28 | **82.5%** (77.1% to 87.5%) | **100%** | **84.5%** (79.5% to 89.6%) | **100%** | 20% |
| 30 | **84.5%** (79.8% to 89.6%) | **100%** | **85.2%** (80.8% to 89.9%) | **100%** | 26% |
| 35 | **88.2%** (84.8% to 92.3%) | **100%** | **88.6%** (85.5% to 91.6%) | **100%** | 50% |
| 38 | **89.2%** (86.9% to 91.6%) | **100%** | **91.6%** (89.2% to 93.9%) | **100%** | 61% |
| 42 | **92.3%** (90.2% to 94.3%) | **100%** | **92.6%** (90.2% to 94.9%) | **100%** | 75% |

week 42, because cases kept arriving afterward; in those, the lead time per season is a minimum.

### C. Without the publication lag

Without the 6-week lag, with the data as they are in the notification system on each date, everything happens about 6 weeks earlier, but the lead time barely changes (Figure 4). M exceeds 75% in week 11 (strict rule, 20) and A in week 4 (strict rule, 18); half of the high seasons become evident in week 29. The lead time is 18 weeks for M (CI 11 to 24; 9 under the strict rule) and 25 for A (CI 14 to 30; 11 under the strict rule). The composition chosen for M is again projection and off-season cases, and M starts saying "high" before the peak in 87% of the high seasons.

### D. Sensitivity analyses

The composition chosen for M was projection and off-season cases in all analyses. The lead time of M stayed between 17

Table VI. Performance in the start week, under the main and strict rules: overall accuracy, sensitivity (high seasons identified), specificity (non-high seasons identified) and range of accuracy across the 11 seasons.

| Criterion and rule | Week | Accuracy | Sensitivity | Specificity | Accuracy per season |
|---|---|---|---|---|---|
| M, main rule | 18 | 75.1% | 75.7% | 74.5% | 59% to 96% |
| M, strict rule | 26 | 81.1% | 76.5% | 85.1% | 67% to 93% |
| A, main rule | 10 | 76.1% | 70.6% | 80.7% | 63% to 85% |
| A, strict rule | 24 | 82.8% | 76.5% | 88.2% | 70% to 96% |

Table VII. Composition of M, start under the main rule, evidence week (half of the high seasons evident, or 25% and 75% in the last two rows) and lead time with 95% CI (resamples without a window counted as zero), for criterion M and projection A. Weeks of the season; "after 42" indicates that evidence was not reached by the last week evaluated, and the lead time is then a minimum.

| Analysis | Composition of M | Start M | Start A | Evidence | Lead time M | Lead time A |
|---|---|---|---|---|---|---|
| Main (6-week lag) | projection + off-season cases | 18 | 10 | 35 | 17 (10 to 25) | 25 (14 to 30) |
| No lag | projection + off-season cases | 11 | 4 | 29 | 18 (11 to 24) | 25 (14 to 30) |
| 10-week lag | projection + off-season cases | 21 | 14 | 39 | 18 (11 to 26) | 25 (14 to 30) |
| Without atypical seasons | projection + off-season cases | 25 | 13 | 36 | 11 (1 to 25) | 23 (6 to 27) |
| National season (July to June) | projection + off-season cases | 20 | 9 | after 42 | $\geq$ 23 (6 to 35) | $\geq$ 34 (4 to 35) |
| Evidence with 25% of high seasons | projection + off-season cases | 18 | 10 | 30 | 12 (5 to 19) | 20 (8 to 23) |
| Evidence with 75% of high seasons | projection + off-season cases | 18 | 10 | 42 | 24 (16 to 32) | 32 (19 to 35) |

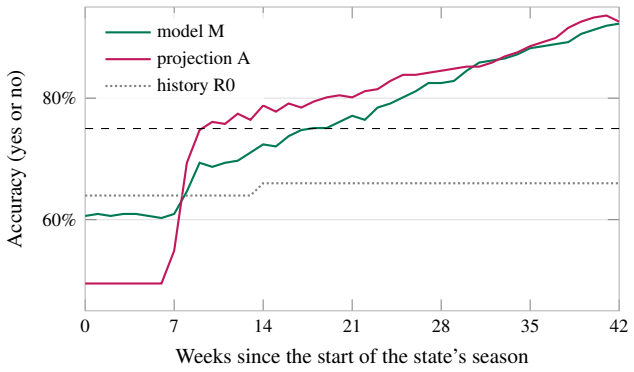



Fig. 3. Accuracy by week of the season with data from the open file (6-week lag): criterion M, projection A and history R0. The dashed line is the 75% threshold.

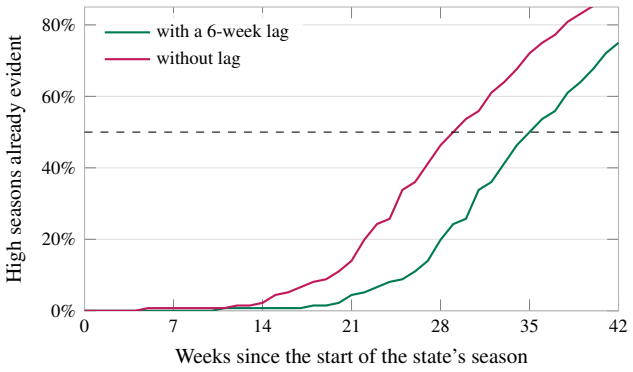



Fig. 4. Fraction of the high seasons already evident from the count (probable cases available above 300 per 100,000) in each week of the season, with the 6-week lag of the open file and without lag. The dashed line marks half.

and 18 weeks with 0, 6 and 10 weeks of lag, and that of A at 25; what changes with the lag is the week in which everything happens. Without the atypical seasons, also excluded from the calculation of the predictors and the history, M only exceeds 75% in week 25, and the lead time drops to 11 weeks, with an interval of 1 to 25 and 1.6% of the resamples without a window; A keeps 23. The measure depends on how much is required of the count: with 25% of the high seasons evident, M anticipates by 12 weeks; with 75%, by 24. With the national season, from July to June, half of the high seasons do not become evident by week 42 of the season, and the lead times become minimums (23 weeks or more for M and 34 or more for A); in this version too, projection and off-season cases was the best pair, with a gain over the projection alone in 99% of the resamples.

### E. Registered forecast for 2026/27

With the files downloaded on 7 Nov 2026 and the seasons with a known outcome on that date, the models produced the forecast in Table IX, dated 1 Nov 2026 (data up to 20 Sep 2026), for the 24 states whose season had already begun. In Alagoas, Paraíba and Maranhão, the season had not yet begun. The forecast was recorded by commit before any later data were downloaded and will be checked when there is a file covering the whole season. In these early weeks (2 to 17), the historical accuracy of M lies between 60% and 75%; the table is a record for checking, and not a reliable forecast.

M gives 50% or more in 14 of the 24 states, against 10 for A and 13 for R0. In several states, the season was in weeks 2 to 6 and the lag had not yet let cases from the season arrive; there, A has no information of its own and M stays close to the

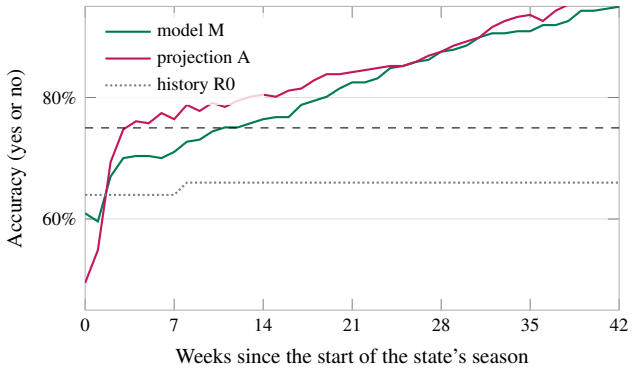



Fig. 5. Accuracy by week of the season without publication lag: criterion M, projection A and history R0. The dashed line is the 75% threshold. The comparison with Figure 3 shows the effect of the 6-week lag of the open file.

Table VIII. Probability of a high-incidence season in 2026/27, by state, on 1 Nov 2026: week of the season on that date, history R0, projection A and criterion M (projection and off-season cases). A record for checking, not a warning.

| State | Week | R0 | A | M |
|---|---|---|---|---|
| GO | 9 | 95% | 94% | 98% |
| AC | 11 | 75% | 82% | 82% |
| MS | 12 | 70% | 84% | 79% |
| MT | 9 | 65% | 81% | 75% |
| SP | 8 | 65% | 65% | 73% |
| TO | 8 | 55% | 77% | 73% |
| MG | 8 | 55% | 83% | 70% |
| ES | 6 | 70% | 46% | 70% |
| DF | 6 | 65% | 46% | 65% |
| PR | 5 | 60% | 46% | 60% |
| RN | 3 | 60% | 46% | 60% |
| RO | 9 | 45% | 75% | 58% |
| CE | 3 | 50% | 46% | 50% |
| BA | 5 | 50% | 46% | 50% |
| RJ | 4 | 40% | 46% | 40% |
| SC | 8 | 25% | 96% | 37% |
| PE | 2 | 35% | 46% | 35% |
| RR | 3 | 30% | 46% | 30% |
| AP | 16 | 33% | 24% | 30% |
| RS | 17 | 24% | 24% | 27% |
| PI | 2 | 25% | 46% | 25% |
| AM | 9 | 15% | 50% | 18% |
| PA | 15 | 10% | 46% | 13% |
| SE | 3 | 10% | 46% | 10% |

history.

## V. DISCUSSION

### A. Main findings

With the open SINAN file, the season defined by state and only outcomes known at each date, criterion M, made up of the count projection and off-season cases, forecasts a high season with average accuracy of 75% or more from week 18, 17 weeks before the count makes half of the high seasons evident (9 under the strict rule). Off-season cases were the predictor that added the most to the projection, with a gain in all resamples. The projection alone (A), however, reached 75% earlier, with 25 weeks of lead time (11 under the strict rule), and was as accurate as M or more in almost every week: procedure M showed no clear advantage over A.

### B. Promising aspects and limits, side by side

**On the promising side**, with a chance of 75% or more:

- the window of M and that of A existed in all resamples, with lead times of 17 and 25 weeks under the main rule;
- the accuracy of M of 78.5% or more from week 23 and that of A of 80% or more from week 19 (chances above 84%);
- the gain of off-season cases over the projection alone, in the same model (chance of 100%), and that of the pair over A (chance of 94%).

**On the negative side**, at the start under the main rule, accuracy has an interval that goes down to 69%, and the chance of M exceeding 75% in that week is 52%; under the strict rule, the lead time drops to 9 and 11 weeks. Average accuracy of 75% is not 75% in each season: there were seasons with 59% (M) and 63% (A) at the start. Sensitivity at the start was between 71% and 76%, and false alarm between 19% and 26%. Procedure M changed models over time and only started choosing off-season cases in the last three seasons.

These highlights come from curves read over 43 weeks and in several analyses; some of them could appear by chance. For that reason the note only counts as the start the point at which accuracy stays stably above 75%, reports the strict rule, and isolated numbers should be read as a signal, not as proof.

### C. Comparison with the literature

The pattern is consistent with what other studies have found. In the open challenge in the Americas, the models with more biology were not better than the simple ones [5]; here, no climate signal alone improved on the history, and the best complement to the projection was another signal from the count itself, off-season cases. In Vietnam, the gain over the historical baseline existed at 1 to 3 months ahead, but not at 4 to 6 [6]; here, the lead time is measured against a detection rule based on cumulative incidence, and not against the history. The importance of the reporting delay is what motivates InfoDengue's nowcasting [2] and the studies with digital data [9].

### D. Strengths

- Lead time measured against the count of probable cases itself, with two start rules and with the fraction of sea-

sons anticipated before the peak.

- Season defined by state and a series of probable cases only, the same in all years.
- Training, history and criterion selection cut off by the date on which each outcome became known.
- Screening of six predictors alone and in pairs, with the incremental gain of each one measured in the same model.
- All 27 states, 11 seasons and 43 weeks, with uncertainty that respects the dependence between states in the same year; registered prospective forecast; individual forecasts, code and aggregated data open.

### E. Weaknesses of the method

- **Approximate reconstruction of the past.** The final classification of the cases, the current climate series and the delay distribution from other years are not exactly what was seen at each date. The likely effect is to make the count cleaner than it was in real time.
- **Low statistical power.** With 11 seasons, the interval of the lead time of M runs from 10 to 25 weeks.
- **Yes-or-no outcome.** The fixed threshold of 300 per 100,000 treats a season of 310 and one of 1,000 the same way.
- **Unit too large.** Epidemics start in cities; the state average dilutes local signals.
- **Trough estimated with 7 years.** The start week of each state was calculated with 2007 to 2013 and may have changed.
- **Conservative training.** The training for each week uses the outcomes known at the date of the state that starts earliest, which removes information from the others.
- **Constructed lead time.** The main measure compares two aggregate curves; it is not the lead time relative to the start of the epidemic or to the need to expand care.
- **Model M anchored in the history.** M tends to say "high" early in the states that usually have a high season, which increases the lead time per season and also the false alarm rate.
- **Notification is not infection.** Changes in care seeking, testing and notification, including with the arrival of the vaccine, alter the outcome without changing transmission.

### F. What the result allows us to conclude

It allows us to conclude that, in this retrospective reconstruction, public data anticipate with average accuracy of 75% which states will have a high season several weeks before the count of probable cases makes it evident, and that off-season cases are the predictor that adds the most to the count projection. It does not allow the weeks of lead time to be treated as demonstrated operational capacity: the number depends on the start rule, the reconstruction of the past is approximate and only the prospective check will tell how much of it holds. Nor does it allow the conclusion that climate does not matter for dengue, or that El Niño has no effect: the signals were measured by state and added to a projection that already carries the effect of climate on cases.

### G. Next steps

Each path below only becomes work with a new protocol: check the forecast registered for 2026/27; redo the analysis with the InfoDengue weekly series [2], through the Mosqlimate API [20], which includes notifications not yet classified; test off-season cases with other definitions of the low season; and go down to health regions or large municipalities.

## VI. CONCLUSION

With criterion M, made up of the case count projection and off-season cases, a high season can be forecast 17 weeks ahead of the simple case count (95% CI 10 to 25 weeks), with average accuracy of 75% or more from week 18 of the state's season; under the strict rule, which requires the lower limit of the interval to be above 75%, the lead time is 9 weeks. Off-season cases were the predictor that added the most to the count projection. The projection alone, however, reaches 25 weeks (11 under the strict rule) and was as accurate as M or more in almost every week, so procedure M showed no clear advantage over it. These numbers come from an approximate retrospective reconstruction of what was known at each date and are not yet demonstrated operational capacity: checking the forecast registered for 2026/27 is the next test.

## VII. LIMITATIONS

- The research steps (collection, database assembly, calculation of the predictors, models, analysis and writing) were carried out with artificial intelligence and did not undergo independent human validation. The results may contain errors or gaps because of this.
- The design weaknesses listed in the Discussion also apply as limitations.
- Registering the protocol by commit documents the order between protocol and results, but does not prove that

the data had not been explored before; the retrospective evaluation is not blind.

- Dengue vaccination and circulating serotypes were not included in the models.
- The forecast for 2026/27 does not cover Alagoas, Paraíba and Maranhão.

## VIII. GLOSSARY

Technical terms used in this note, with attention to those that are not common in traditional epidemiology studies.

- **95% confidence interval:** range of values compatible with the data.
- **Accuracy:** fraction of state-season pairs in which the yes-or-no forecast matched what happened.
- **Bootstrap:** redoing the calculation many times by drawing the data with replacement, to measure how much the result varies by chance.
- **Brier and BSS:** mean squared error of the probability; the BSS is the fraction of that error that one model eliminates relative to another.
- **Censoring:** when the expected event does not happen by the end of follow-up; the measured value is then a minimum.
- **Chance of accuracy of 75% or more:** fraction of resamples in which accuracy was 75% or more; from 75% upward, the note calls it promising.
- **Criterion M:** the screening model that best forecast the past seasons, chosen without looking at the season being forecast.
- **El Niño and ONI:** periodic warming of the Pacific Ocean that alters the climate in Brazil; the ONI is the index that measures this warming.
- **Epidemiological week:** Sunday-to-Saturday week used by surveillance.
- **Evident from the count:** a season whose probable cases already available have exceeded 300 per 100,000.
- **Expanding window:** scheme in which each season is forecast using only the previous ones.
- **False alarm:** a season that was not high and that the model was calling “high”.
- **Imputation:** estimating a missing value from similar data; here, the data entry date from 2016 to 2020.
- **Incremental gain:** how much one predictor improves another within the same model.
- **Known outcome:** result of a past season treated as available; here, 8 weeks after the end of the season plus the publication lag.
- **Lead time:** number of weeks between the model starting to be right 75% or more of the time and the count making half of the high seasons evident.
- **Logistic regression:** model that turns signals into a probability between 0 and 1.
- **Mild winter:** a winter warmer than the state’s average, measured over the 91 normally coldest days.
- **Monte Carlo error:** imprecision of a number calculated by random draws; it decreases with more draws.
- **Nested selection:** choosing the model using only the seasons before the one to be forecast.
- **Nowcasting:** estimating the current number of cases by correcting for those not yet reported.
- **Off-season cases:** incidence in the state’s low season above or below the normal of previous seasons.
- **Offset:** fixed term of a model; here, the state’s history, from which the predictors can only move the forecast away.
- **Penalization:** rule that shrinks a model’s weights to keep it from learning noise.
- **Probable case:** dengue notification that was not discarded during investigation.
- **Publication lag:** time between a case entering the notification system and appearing in the open file; here, about 6 weeks.
- **Reanalysis:** gridded reconstruction of past climate, combining observations and physical models; here, MERRA-2.
- **Screening:** comparison of each predictor alone and in pairs.
- **Sensitivity:** fraction of the high seasons that the model called “high”.
- **Specificity:** fraction of the non-high seasons that the model called “not high”.
- **Strict rule:** start requiring the lower limit of the 95% interval of accuracy to exceed 75%.
- **Thermal suitability:** number from 0 to 1 that indicates how much temperature favors dengue transmission by *Aedes aegypti*; here, an approximation that is zero in extreme cold and heat and peaks at 29.1 °C.
- **Trough week:** epidemiological week with the lowest mean dengue incidence in a state; here, the start of that state’s season.

## IX. REPORTING GUIDELINES

The note follows, insofar as they apply to population-level forecasting, the TRIPOD+AI [12], EPIFORGE [13] and RECORD [14] guidelines. The main items are in the sections: context and questions (I), sources, period, probable case series, season by state, outcome and information available at each date (II), design, predictors, screening, criterion M, measures, uncertainty and criteria (III), screening, performance with uncertainty, sensitivity and specificity, lead time and sensitivity analyses (IV.A to IV.D), registered prospective forecast (IV.E), comparison with the literature and weaknesses (V), limitations (VII), use of AI, data and code (form at the end). The individual forecasts by state, season and week are published. There are no individual data on people; all tables are by state.

## DISCLAIMER

This is a working note in health informatics, produced with extensive use of artificial intelligence, and not a clinical study. The research steps carried out with AI (data collection, literature screening, analysis and code) have not undergone independent human validation, and the results may contain errors or gaps as a consequence; the author reviews the final text before publication. It has not been reviewed by a licensed health professional. It does not constitute advice, diagnosis, prognosis or treatment recommendation for any person. Hypotheses are hypotheses, not confirmed findings. The source data have limitations described in the text. Any use in health care requires specific review by licensed professionals, according to the intended use. Valid for the date and version indicated.

## ARTIFICIAL INTELLIGENCE USE STATEMENT

Tools: Claude Opus 5.5 (Anthropic), in a session on 7 Nov 2026. Steps with AI: literature search, protocol design, all code (collection, database, signals, models, evaluation, figures), running the analyses, writing and translation into English and Spanish. Checking: automatic only: the code runs end to end from raw files with SHA-256 hashes; the numbers in the text were checked by script against the result files; DOIs and links were verified. There was no independent human validation of the AI steps, and the results may contain errors or gaps because of this; the author reviewed the final text. Tool limitations: it can make mistakes in code and in reading the literature that automatic checks do not catch. Record: protocol, code and results versioned in the program's public repository (github.com/willianwistuba/mais-saude-ai), with the protocol in commit a9c960e and the 2026/27 forecast in commit dba78f5. Conflicts and funding: none.

## DATA AND CODE AVAILABILITY

All data used are public, and the scripts that produce every number are available at https://mais-saude-ai.vercel.app/en/code/, with instructions to reproduce the results.

## ETHICS

This note uses only public, aggregated data that do not identify individuals, which waives ethics committee review under Brazilian National Health Council Resolution 510/2016 and CONEP Circular Letter 17/2022. Counts below five are suppressed at municipal level, and no re-identification is attempted.

## COMPETING INTERESTS AND FUNDING

The author declares no competing interests. This work received no funding, and the project accepts no resources from manufacturers of technologies it assesses or comments on. This text has not been peer reviewed, reviewed by a health professional or subjected to independent human validation of the research steps carried out with AI (the author reviews only the final text), and should be read as a working note. Corrections create a new version, and the previous one remains accessible.

## HOW TO CITE

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